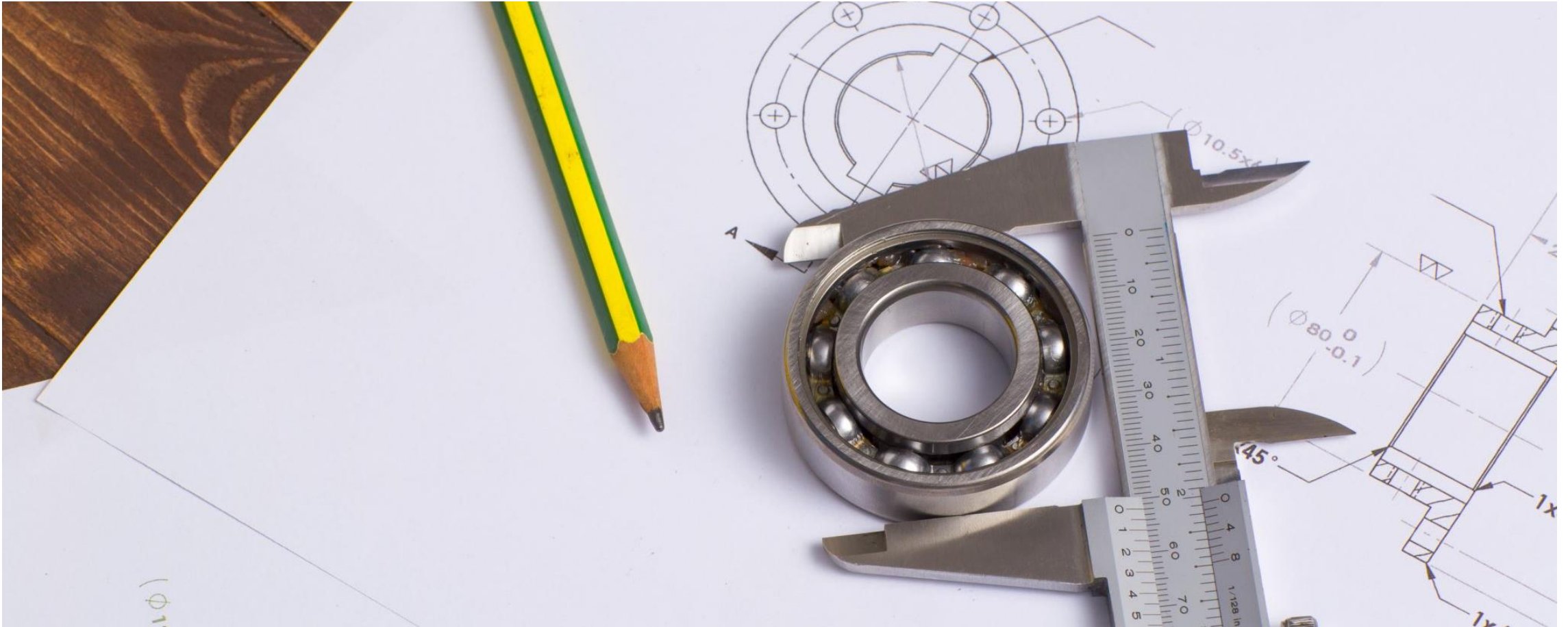


Mechanical Design Lifting Assembly Task

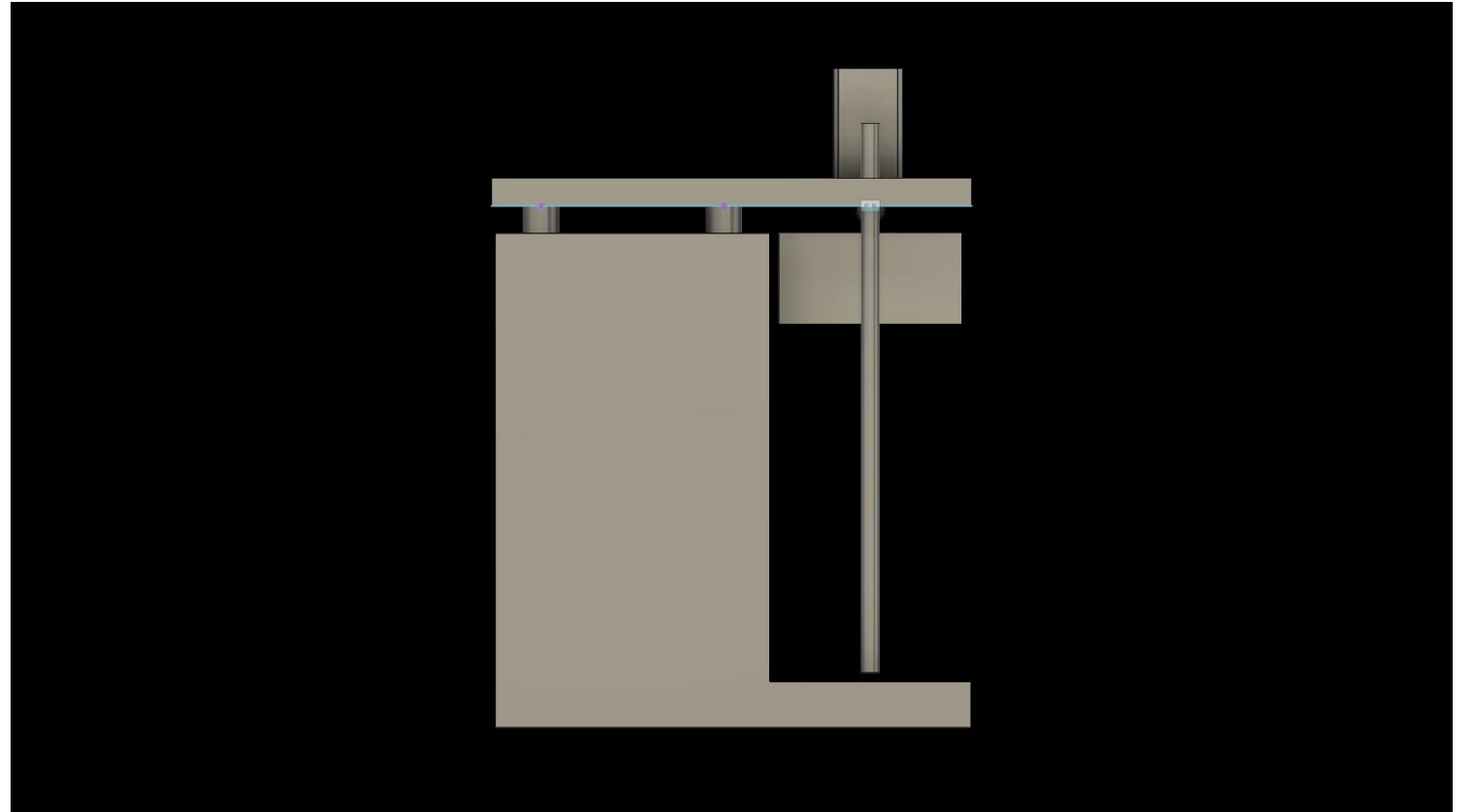


Prepared by Ravi Teja Bollineni

Design Overview & Strategy

Key Points:

- Mounting: Securely interfaced with the four M10 bosses on the base frame.
- Cantilever Design: 15 mm steel bridge plate engineered to position the lift axis precisely over the payload zone.
- Stability: Dual 20 mm guide rails prevent “racking” or twisting of the heavy 31.6 kg load during high-speed travel.



Performance & Calculations

- Target: Lift Speed ≥ 100 mm/s
- Component Selection: Motor A (50W) + 1605 Lead Screw (5 mm pitch).
- Formula: $V = (\text{RPM} * \text{Pitch})/60$
- Result = $(1200 \text{ RPM} * 5 \text{ mm})/60 = 100 \text{ mm/s}$
- Stroke: Total rod length of 600 mm accommodates the required 500 mm lift height with safety margins.
- Force required: $F = 31.6 \text{ kg} * 9.81 \sim 310 \text{ N}$
- Estimated Torque: For a 5 mm pitch screw, the required torque is roughly 0.4 to 0.6 Nm (including friction).
- Safety Margin: Since Motor A (50W) typically provides enough torque for this range, selection is solid.



Verification & Testing Plan

Digital Testing: Use FEA (Finite Element Analysis) to ensure the cantilevered plate maintains a high safety factor (>2.0) under the 31.6 kg load.

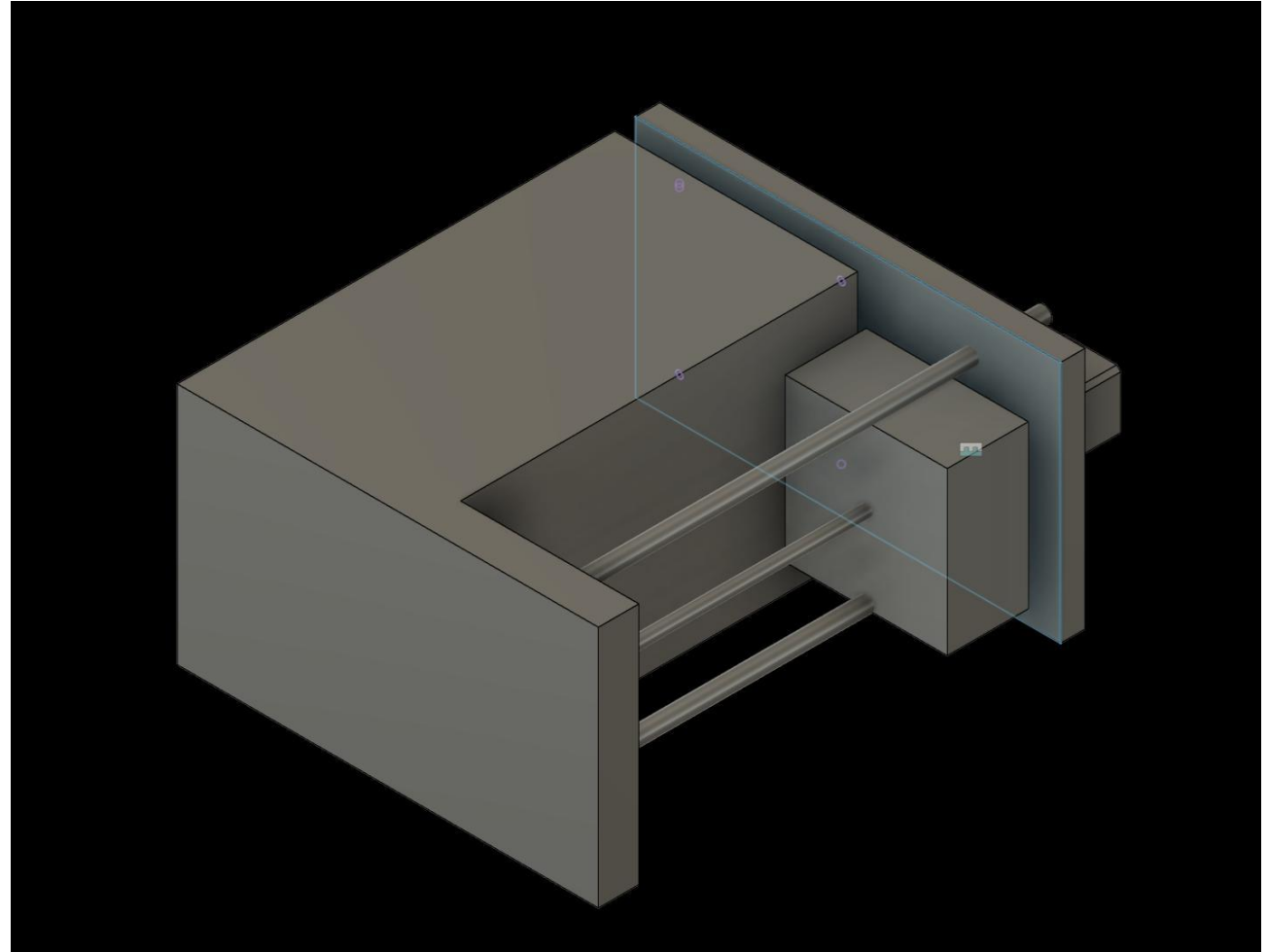
Physical Testing:

- Static Load Test: Verify zero "back-drive" or creep when the motor is unpowered.
- Dynamic Cycle Test: 500+ cycles at 100 mm/s to monitor heat and rail alignment.
- Precision Check: Using a laser tachometer to verify consistent 1200 RPM under full load.

Conclusion & Files

The design successfully meets all Millitec performance targets using cost-effective, standard industrial components (Motor A, 1605 Ball Screw, 20 mm Linear Rails).

*Note: Ravi Teja Millitec Final Assembly.step file is attached for further technical review.



THANK YOU